

Linear Algebra

Vectors, Matrices, Linear Transformations and Vector Spaces

Asnifa P.K.

B.Tech Computer Science and Engineering

September 22, 2026

Contents

- 1 Vectors and Coordinate Systems
- 2 Vector Addition and Scalar Multiplication
- 3 Linear Combinations
- 4 Span and Basis
- 5 Matrix Multiplication
- 6 Matrix Multiplication as Composition
- 7 Three-Dimensional Linear Transformations
- 8 Determinant
- 9 Inverse Matrices
- 10 Column Space and Null Space
- 11 Nonsquare Matrices
- 12 Dot Product and Duality
- 13 Cross Product
- 14 Cross Product as a Linear Transformation
- 15 Cramer's Rule
- 16 Change of Basis

What is a Vector?

Definition

A vector is a mathematical quantity that has both **magnitude** and **direction**.

A vector in 2D can be written as

$$\vec{v} = \begin{bmatrix} x \\ y \end{bmatrix}$$

and a vector in 3D as

$$\vec{v} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Example:

$$\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Magnitude:

$$|\vec{v}| = \sqrt{3^2 + 4^2} = 5$$

Coordinate System

A coordinate system gives a reference framework for locating points and representing vectors. For 3D space, the standard coordinate axes are:

$$x, \quad y, \quad z$$

The standard basis vectors are

$$\hat{i} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \hat{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \hat{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore,

$$\vec{v} = x\hat{i} + y\hat{j} + z\hat{k}.$$

Example:

$$\vec{v} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = 2\hat{i} + 3\hat{j} + 4\hat{k}.$$

Vector Addition

Vectors are added component by component.

Let

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}.$$

Then

$$\vec{u} + \vec{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \end{bmatrix}.$$

Example:

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$\vec{u} + \vec{v} = \begin{bmatrix} 1 + 4 \\ 2 + 5 \\ 3 + 6 \end{bmatrix} = \boxed{\begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}}$$

Scalar Multiplication

A scalar is an ordinary number.

Multiplying a vector by a scalar means multiplying every component by that number.

$$k \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} kx \\ ky \\ kz \end{bmatrix}.$$

Example:

$$3 \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \\ 12 \end{bmatrix}.$$

If $k > 1$, the vector is stretched.

If $0 < k < 1$, the vector is shortened.

If $k < 0$, its direction is reversed.

Linear Combination

A linear combination is formed by multiplying vectors by scalars and adding the results.

$$\vec{v} = c_1 \vec{v}_1 + c_2 \vec{v}_2 + \cdots + c_n \vec{v}_n.$$

Here,

$$c_1, c_2, \dots, c_n$$

are called **scalar coefficients**.

Example:

Let

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Find

$$2\vec{v}_1 + 3\vec{v}_2.$$

$$= 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 9 \\ 3 \end{bmatrix} = \boxed{\begin{bmatrix} 11 \\ 7 \end{bmatrix}}$$

Span

The **span** of a set of vectors is the collection of all possible linear combinations of those vectors.

$$\text{Span}\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n\}.$$

For example,

$$\text{Span}\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\} = \mathbb{R}^2.$$

Why?

Any vector

$$\begin{bmatrix} x \\ y \end{bmatrix}$$

can be written as

$$x \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Therefore, these two vectors can generate every vector in $\mathbb{R}^2$.

Basis Vectors

A set of vectors forms a **basis** if:

- 1 The vectors span the space.
- 2 The vectors are linearly independent.

The standard basis for $\mathbb{R}^3$ is

$$\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \vec{e}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

For example,

$$\begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = 2\vec{e}_1 + 3\vec{e}_2 + 4\vec{e}_3.$$

Matrices

A matrix is a rectangular arrangement of numbers.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

The matrix above has:

2 rows $\times$ 2 columns.

Matrices can be used to represent:

- Systems of equations
- Linear transformations
- Rotations
- Scaling
- Reflections

Matrix Multiplication

Matrix multiplication is performed using the **row-by-column rule**.

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 1(5) + 2(7) & 1(6) + 2(8) \\ 3(5) + 4(7) & 3(6) + 4(8) \end{bmatrix}$$
$$= \boxed{\begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}}$$

Matrix Multiplication as Composition

A matrix can represent a transformation.

Suppose

$$T(\vec{x}) = A\vec{x}$$

and

$$S(\vec{x}) = B\vec{x}.$$

If we first apply T and then S :

$$S(T(\vec{x})) = B(A\vec{x})$$

Therefore,

$$S \circ T \longleftrightarrow BA.$$

Important

The matrix on the right acts first.

So,

$$BA\vec{x}$$

means:

$$\boxed{\vec{x} \xrightarrow{A} A\vec{x} \xrightarrow{B} BA\vec{x}}$$

Linear Transformation

A transformation T is linear if

$$T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$$

and

$$T(c\vec{u}) = cT(\vec{u}).$$

Every matrix A defines a linear transformation:

$$T(\vec{x}) = A\vec{x}.$$

For example,

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

gives

$$T \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ 3y \end{bmatrix}.$$

This stretches the x and y directions by different amounts.

Three-Dimensional Transformations

A 3×3 matrix transforms vectors in $\mathbb{R}^3$.

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}.$$

For

$$\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$

the transformed vector is

$$T(\vec{x}) = A\vec{x}.$$

Determinant

For a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is

$$\det(A) = ad - bc.$$

Example:

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}.$$

Therefore,

$$\det(A) = 2(4) - 3(1)$$

$$= 8 - 3$$

$$\boxed{= 5}.$$

Since

$$\det(A) \neq 0,$$

the matrix is invertible.

Geometric Meaning of Determinant

The determinant tells us how a linear transformation changes area or volume.
For a 2D transformation:

$$|\det(A)|$$

represents the area scaling factor.

If

$$\det(A) = 2,$$

the area is doubled.

If

$$\det(A) = \frac{1}{2},$$

the area becomes half.

If

$$\det(A) = 0,$$

the transformation collapses the space into a lower dimension.

Important

$$\det(A) = 0 \implies A^{-1} \text{ does not exist.}$$

Inverse Matrix

The inverse of a matrix A is written as

$$A^{-1}.$$

It satisfies

$$AA^{-1} = A^{-1}A = I.$$

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

The inverse exists only if

$$ad - bc \neq 0.$$

Example: Inverse Matrix

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

First find the determinant:

$$\det(A) = 2(1) - 1(1) = 1.$$

Therefore,

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

Check:

$$AA^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Thus,

$$\boxed{AA^{-1} = I}.$$

Column Space

The column space of a matrix is the span of its columns.

Let

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

Its columns are

$$\vec{c}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \vec{c}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Since

$$\vec{c}_2 = 2\vec{c}_1,$$

only one independent direction exists.

Therefore,

$$\text{Col}(A) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}.$$

Meaning

Column space represents all possible outputs of $A\vec{x}$.

Null Space

The null space contains all vectors $\vec{x}$ satisfying

$$A\vec{x} = 0.$$

Consider

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

Let

$$\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix}.$$

Then

$$A\vec{x} = 0$$

gives

$$x + 2y = 0.$$

Therefore,

$$x = -2y.$$

Let $y = t$:

$$\vec{x} = \begin{bmatrix} -2t \\ t \end{bmatrix} = t \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Thus,

$$\text{Null}(A) = \text{Span} \left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}$$

Nonsquare Matrices as Transformations

A nonsquare matrix can transform vectors between different dimensions.
Consider

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \end{bmatrix}.$$

This is a 2×3 matrix.

Therefore,

$$A : \mathbb{R}^3 \rightarrow \mathbb{R}^2.$$

For

$$\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix},$$

we get

$$A\vec{x} = \begin{bmatrix} x + 2z \\ y + 3z \end{bmatrix}.$$

So a 3D vector is transformed into a 2D vector.

Dot Product

For two vectors

$$\vec{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix},$$

the dot product is

$$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

It can also be written as

$$\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta.$$

Example:

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}.$$

$$\vec{u} \cdot \vec{v} = 1(4) + 2(-1) + 3(2) = 4 - 2 + 6 = \boxed{8}.$$

Dot Product and Orthogonality

If

$$\vec{u} \cdot \vec{v} = 0,$$

then

$$\vec{u} \perp \vec{v}.$$

Example:

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Then

$$\vec{u} \cdot \vec{v} = 1(2) + 2(-1) = 2 - 2 = 0.$$

Therefore,

$$\boxed{\vec{u} \perp \vec{v}}.$$

Duality

The dot product can be viewed as a function that takes a vector and produces a scalar.
For a fixed vector $\vec{a}$:

$$f(\vec{x}) = \vec{a} \cdot \vec{x}.$$

For example, let

$$\vec{a} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Then

$$f \begin{bmatrix} x \\ y \end{bmatrix} = 2x + 3y.$$

This is a linear functional.

Idea

Duality connects vectors with linear functions that map vectors to numbers.

Cross Product

The cross product is defined for vectors in $\mathbb{R}^3$.

$$\vec{u} \times \vec{v}$$

produces a vector perpendicular to both $\vec{u}$ and $\vec{v}$.

For

$$\vec{u} = (u_1, u_2, u_3), \quad \vec{v} = (v_1, v_2, v_3),$$

$$\vec{u} \times \vec{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}.$$

Cross Product: Example

Let

$$\vec{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}.$$

$$\vec{u} \times \vec{v} = \begin{bmatrix} 2(6) - 3(5) \\ 3(4) - 1(6) \\ 1(5) - 2(4) \end{bmatrix}$$

$$= \begin{bmatrix} 12 - 15 \\ 12 - 6 \\ 5 - 8 \end{bmatrix}$$

$$= \begin{bmatrix} -3 \\ 6 \\ -3 \end{bmatrix}$$

This vector is perpendicular to both $\vec{u}$ and $\vec{v}$.

Geometric Meaning of Cross Product

The magnitude is

$$|\vec{u} \times \vec{v}| = |\vec{u}||\vec{v}| \sin \theta.$$

This represents the area of the parallelogram formed by $\vec{u}$ and $\vec{v}$.

The area of the triangle is

$$\boxed{\frac{1}{2}|\vec{u} \times \vec{v}|}.$$

If the vectors are parallel,

$$\theta = 0^\circ$$

and therefore

$$\vec{u} \times \vec{v} = 0.$$

Cross Product as a Linear Transformation

Fix a vector

$$\vec{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}.$$

Define

$$T(\vec{x}) = \vec{a} \times \vec{x}.$$

This operation can be represented using a matrix:

$$T(\vec{x}) = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \vec{x}.$$

Therefore, taking the cross product with a fixed vector is a linear transformation.

Cramer's Rule

Cramer's rule is a method for solving systems of linear equations.

Consider

$$ax + by = e$$

$$cx + dy = f.$$

The determinant is

$$D = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

If

$$D \neq 0,$$

then the system has a unique solution.

Cramer's Rule: Example

Solve:

$$2x + y = 5$$

$$x - y = 1.$$

The coefficient determinant is

$$D = \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -2 - 1 = -3.$$

For x :

$$D_x = \begin{vmatrix} 5 & 1 \\ 1 & -1 \end{vmatrix} = -5 - 1 = -6.$$

Therefore,

$$x = \frac{-6}{-3} = 2.$$

For y :

$$D_y = \begin{vmatrix} 2 & 5 \\ 1 & 1 \end{vmatrix} = 2 - 5 = -3.$$

Therefore,

$$y = \frac{-3}{-3} = 1.$$

$$\boxed{x = 2, \quad y = 1}$$

Change of Basis

The coordinates of a vector depend on the basis being used.

Suppose B and C are two different bases.

Then

$$[\vec{v}]_C = P_{C \leftarrow B} [\vec{v}]_B.$$

Here,

$$P_{C \leftarrow B}$$

is called the change-of-basis matrix.

Important

The actual vector does not change. Only its coordinate representation changes.

Change of Basis: Example

Let the standard basis be

$$E = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}.$$

Consider another basis

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

Suppose

$$[\vec{v}]_B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Then

$$\begin{aligned}\vec{v} &= 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 5 \\ -1 \end{bmatrix}.\end{aligned}$$

Thus,

$$[\vec{v}]_E = \begin{bmatrix} 5 \\ -1 \end{bmatrix}.$$

Eigenvectors and Eigenvalues

For a matrix A , an eigenvector satisfies

$$A\vec{v} = \lambda\vec{v},$$

where

- $\vec{v} \neq 0$ is the eigenvector.
- λ is the eigenvalue.

Rearranging:

$$A\vec{v} - \lambda\vec{v} = 0$$

$$(A - \lambda I)\vec{v} = 0.$$

For a non-zero solution:

$$\det(A - \lambda I) = 0$$

This is the characteristic equation.

Eigenvalue: Example

Consider

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

Find the eigenvalues.

$$A - \lambda I = \begin{bmatrix} 2 - \lambda & 0 \\ 0 & 3 - \lambda \end{bmatrix}.$$

Therefore,

$$\det(A - \lambda I) = (2 - \lambda)(3 - \lambda).$$

Set equal to zero:

$$(2 - \lambda)(3 - \lambda) = 0.$$

Hence,

$$\boxed{\lambda_1 = 2, \quad \lambda_2 = 3}.$$

Finding Eigenvectors

For $\lambda = 2$:

$$(A - 2I)\vec{v} = 0.$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Therefore,

$$y = 0.$$

So,

$$\vec{v} = \begin{bmatrix} x \\ 0 \end{bmatrix}.$$

Choosing $x = 1$:

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

is an eigenvector corresponding to $\lambda = 2$.

Similarly, for $\lambda = 3$:

$$\vec{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Geometric Meaning of Eigenvectors

For a general vector:

$$A\vec{v}$$

may change both its magnitude and direction.

For an eigenvector:

$$A\vec{v} = \lambda\vec{v}.$$

The direction remains the same.

Only the magnitude changes by the factor λ .

Easy Way to Remember

Eigenvector: special direction.

Eigenvalue: scaling factor in that direction.

Vector Spaces

A vector space is a collection of objects that is closed under vector addition and scalar multiplication and satisfies the vector-space axioms.

Examples:

$$\mathbb{R}^2, \quad \mathbb{R}^3$$

Other examples include:

- Polynomial spaces
- Matrix spaces
- Function spaces

For example, the set

$$P_2 = \{a + bx + cx^2 : a, b, c \in \mathbb{R}\}$$

is a vector space.

Dimension of a Vector Space

The dimension of a vector space is the number of vectors in any basis of that space.
For example,

$$\mathbb{R}^2$$

has basis

$$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}.$$

Therefore,

$$\dim(\mathbb{R}^2) = 2.$$

Similarly,

$$\dim(\mathbb{R}^3) = 3.$$

For the polynomial space P_2 ,

$$\{1, x, x^2\}$$

is a basis.

Therefore,

$$\dim(P_2) = 3.$$

Important Relationships

The major concepts can be connected as follows:

Vectors $\rightarrow$ Linear Combinations $\rightarrow$ Span $\rightarrow$ Basis $\rightarrow$ Dimension

Matrices provide a way to represent transformations:

Matrix $\rightarrow$ Linear Transformation

Important matrix concepts:

Determinant $\rightarrow$ Inverse $\rightarrow$ Column Space $\rightarrow$ Null Space

Eigenvalue and Transformation Relationship

A linear transformation can change a vector:

$$\vec{v} \longrightarrow A\vec{v}.$$

For a general vector, both magnitude and direction may change.

For an eigenvector:

$$A\vec{v} = \lambda\vec{v}$$

Therefore, its direction remains unchanged.

Eigenvector = special direction

Eigenvalue = scaling factor

Quick Revision: Formulas

Vector magnitude:

$$|\vec{v}| = \sqrt{x^2 + y^2 + z^2}$$

Dot product:

$$\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$$

Cross product magnitude:

$$|\vec{u} \times \vec{v}| = |\vec{u}| |\vec{v}| \sin \theta$$

2×2 determinant:

$$\det(A) = ad - bc$$

Inverse:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Thank You

Questions?